```
cursor = sqliteconnection.execute("SELECT
id, age, salary from DATABASETABLE")
for row in cursor:
   print ("ID = ", row[0])
   print ("AGE = ", row[1])
   print ("SALARY = ", row[2], "\n")
sqliteconnection.close()
```

Developing applications using Python and SQLite

The interfacing of Python and SQLite can be done in numerous ways. These include GUI based Tkinter programming, Web app development using the Flask framework, and desktop or real-time transaction based applications so that the records can be maintained without any delay as well as with a higher degree of availability and consistency.

Real-time Web based applications using Python and SQLite can be developed and deployed with advanced modules and packages, to help develop secure applications.

In addition to the traditional Web based applications, the integration of advanced libraries can be done for weather forecasting, one time password (OTP) based applications, and many others.

The following example shows how to create a snippet to generate a weather report using the OpenWeatherMap library (*https://openweathermap.org)* with the integration of SQLite and Python. The OpenWeatherMap library provides the application programming interface (API) so that real-time weather predictions and reports can be generated and presented to the client. First, the keys are generated using *https://openweathermap.org/appid#get*, and the following code can be used for the weather report of a specific city or location, with association of the *pyowm* package.

```
import pyowm
owm = pyowm.
OWM('xxxxxxxxxxxxxxxxxxxxxxxx')
observation = owm.weather_at_
place('Fatehgarh Sahib, IN')
w = observation.get_weather()
print(w)
observation_list = owm.weather_around_
coords(-22.57, -43.12)
```

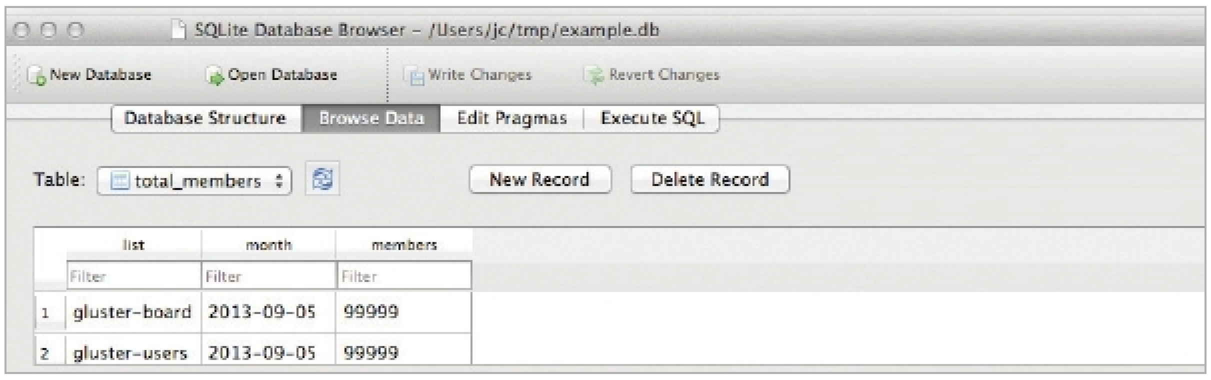

Figure 2: DB Browser for SQLite

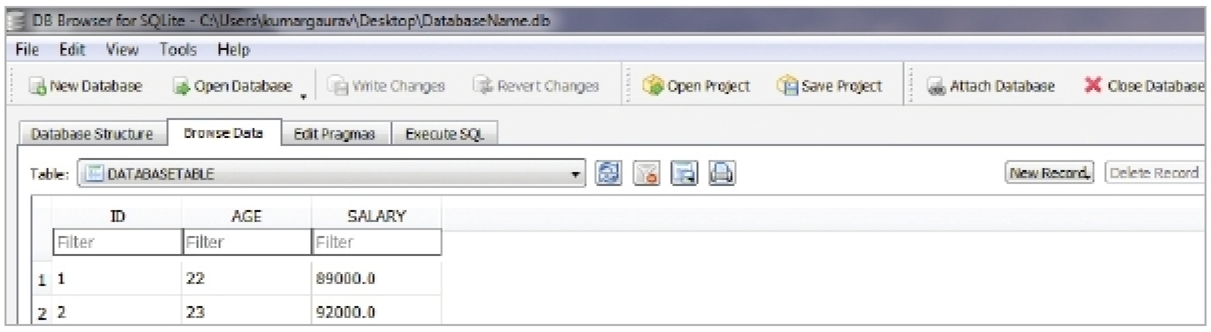

Figure 3: Displaying records in the SQLite browser after inserting records using Python code

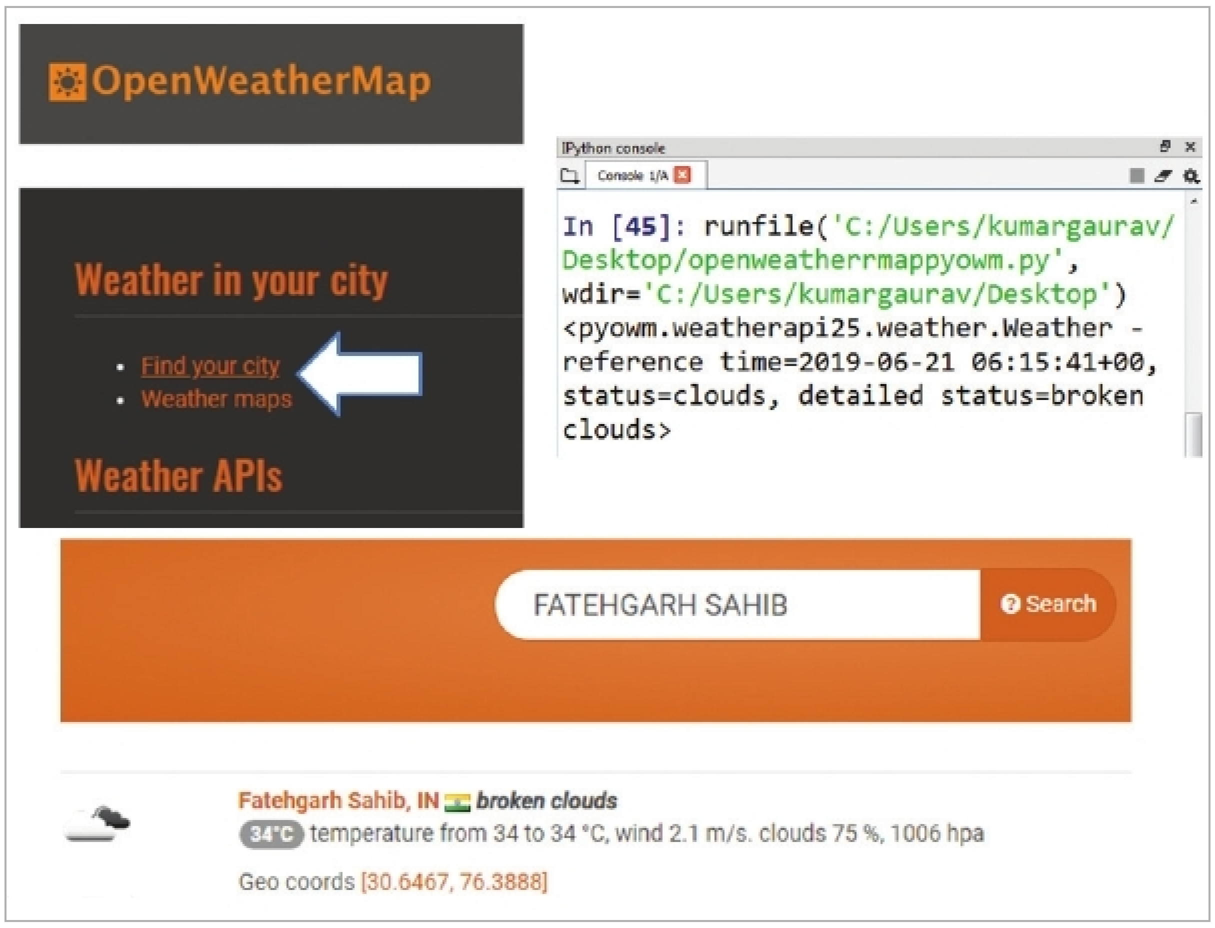

Figure 4: The weather report of the city, Fatehgarh Sahib, in Punjab

On executing the code, the real-time weather is presented; predictive mining can be used in a similar way.

For the purpose of R&D, the performance of SQLite can be compared with other database systems. The code of database connections and retrieval can be implemented on multiple databases including SQLite, CockroachDB, MongoDB, CouchDB, Cassandra, etc. Then the real-time performance in terms of execution time and complexity can be evaluated on the test data sets and records, in order to analyse the effectiveness and metrics of assorted database engines. END

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